MIL-B-14154D
16 May 1985
SUPERSEDING
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24 September 1976

MILITARY SPECIFICATION

BORING MACHINE, CAM AND MAIN BEARING, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of portable, universal boring machine for line boring, facing, and filleting cam and main bearings in motor vehicle engine blocks.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Service.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-889	-	Dissimilar Metals.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206	-	General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02	-	Manual for Machine Tool Gearing.
AGMA 390.03	-	Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Drive, Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI IEEE 268 - Metric Practice.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82	-	Standard Practice for Commercial Packaging.
ASTM E380-84	-	Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions.

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more boring machines for first article inspection (see 4.3 and 6.2.1). The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The boring machine shall be new and of the manufacturer's latest design which shall meet the performance requirements specified herein. The machine shall be an engine block attachment type that is portable and universal. It shall be designed for line boring, facing, and filleting cam and main bearings in motor vehicle engine blocks. The boring bar shall be capable of being driven from either end by an electric drill, not furnished with the machine, or by a handcrank. The design shall be such that all parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the machine or any machine component.

* 3.2.1 Measurement system. The US Customary Systems of Units (US) or the International System of Units (SI) may be used in the design and construction of the boring machine. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84 or ANSI IEEE 268, as applicable.

* 3.2.2 Measuring and indicating devices. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI systems of measurement. When a dual US and SI system is furnished, feed dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as

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specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system (see 3.6.7).

* 3.2.3 Interchangeability. All parts of the machine shall be manufactured to definite standards and tolerances. Parts shall be interchangeable without requiring modification of the part or its mating components. Parts subject to possible future replacement as spare or repair parts shall be assigned a manufacturer's part number and shall be identified in the repair parts manual (see 6.3.1). All replacement, repair, and spare parts shall provide the same or better form, fit, function, and performance as the parts initially installed in the boring machine. When industry or Government standards are applicable, replacement parts shall conform to those standards.

* 3.2.4 Lubrication. All bearings (except sealed-for-life lubricated type), mating gears, and all other moving parts shall be provided with means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the machine tool industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the machine to remove the temporary caps and stoppers before start-up or use of the boring machine.

* 3.2.5 Gears. All gears and pinion designs selected for use in the boring machine or its components shall be manufactured to either English (USA) standard inch diametral pitch per AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio per the International Organization for Standardization (ISO) ISO 54 and ISO 1122/1 and ASTM E380-84.

* 3.2.6 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.2.7 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide access to guarded parts. The boring machine shall comply with the general safety and health requirements promulgated under OSHA 2206 that are applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and

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physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. The boring machine shall be constructed to be capable of performing the operations specified herein. The machine shall be complete so that, when installed and connected to the proper power source, the machine may be used for all operations for which it was designed.

* 3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

* 3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. The methods for preventing loss of tightness or adjustment shall be in accordance with accepted engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

* 3.4.4 Surfaces. Castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

* 3.4.5 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.5 Performance and product characteristics.

3.5.1 Capacity. The boring machine shall perform the operations specified herein, within a bearing diameter range of 1 1/2 through four inches, in engine blocks up to, and including, 60 inches in length, 20 inches in width, and 11 inches in height to bearing center line from the crankcase pan rail surface. The boring machine shall be mounted on the crankcase pan rail surface and shall be capable of being transferred as a unit from one engine block to another for line boring.

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3.5.2 Feed range. A minimum feed range of 0.0035 to 0.0070 inch per revolution of the boring bar shall be provided by the feed mechanism.

* 3.5.3 Accuracy. The boring machine shall be capable of producing finished bearing surfaces that are concentric and without taper to within ±0.0005 inch for each bearing surface and having a surface roughness no greater than 24 microinches arithmetical average (aa). The boring machine shall be capable of line boring the bearing surfaces over the maximum length of the bar, in line, to within 0.003 inch.

3.5.4 Weight. The total weight of the boring machine, including equipment and case, shall not exceed 300 pounds (see 3.6.8).

3.6 Components. The boring machine shall consist of, but not be limited to, the principal components specified in 3.6.1 thru 3.6.8.

3.6.1 Mounting devices. Mounting devices shall be provided to anchor the boring machine securely to the engine block. The devices shall incorporate means for accurate positioning of the boring bar relative to the mean center of the cam and main bearings. All tools and fixtures required for the setup or operation of the boring machine on engine blocks within the specified capacity shall be provided with each machine.

3.6.2 Boring bar. The boring bar shall have internal tool bit holding provisions located along the bar that permit adjustment of the tool bit at the bearing to be finished. Eight lock screws for securing tool bits in boring bars shall be furnished with each boring machine.

3.6.3 Tool bits. Not less than four high speed tool bits shall be provided with each machine for each boring, facing, and filleting operation. The tool bits shall be ground for their intended use and have a Rockwell hardness of 63 to 65 on the C-scale. Unless otherwise specified (see 6.2.1), four tool bits with carbide tips shall be furnished with each machine for the boring of cast iron bearing saddles.

3.6.4 Feed mechanism. The feed mechanism shall provide not less than two feeds covering the feed range specified in 3.5.2. The feed mechanism shall operate both manually and automatically.

3.6.5 Handcrank. A handcrank shall be provided with each machine to manually turn the boring bar (see 3.2).

3.6.6 Flexible coupling. A detachable flexible coupling shall be provided to transfer the power from a 1/2 inch electric drill, not furnished with the machine, to the boring machine (see 3.2).

3.6.7 Measuring devices. A direct reading micrometer for measuring bearing bore diameters within a range of 1 1/4 to four inches, readable to 0.0001 inch, shall be included. All gages, special micrometers, and dial indicators required for the normal setup and operation of the boring machine shall be provided with each machine. All measuring devices furnished shall be permanently marked with clear, bold face characters that can be easily read (see 3.2.2).

3.6.8 Carrying case. The case shall be made of wood or steel designed to contain, protect, and carry the boring machine with its component parts,

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equipment, and tools. The carrying case shall be fitted with a hinged lid with positive acting latches. The carrying case shall have a carrying handle(s) suitable for operator handling or sling handling (see 3.5.4).

* 3.7 Product identification. The name-plate and the information-plate may be incorporated into one or more corrosion resistant metallic plate(s) at the manufacturer's option. The plates shall be in accordance with MIL-STD-130. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The nameplate shall be secured to the machine with permanent fasteners such as screws, bolts, and rivets at each of the four corners. The nameplate shall include the following information:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number

* 3.8 Workmanship. Standards of workmanship shall assure that the boring machine shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

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(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Unless otherwise specified (see 6.2.1), sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test		Inspection Level	AQL
Product examination	(4.5)	II	1.0
Performance test	(4.6.1)	II	1.0
Accuracy test	(4.6.2)	II	1.0
Packaging inspection	(4.7)	S-4	4.0

* 4.5 Examination. Visually and manually examine the boring machine to determine conformance with the requirements of 3.2 through 3.4.5 and 3.5.4 through 3.8. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the machine and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Performance test. The boring machine shall be assembled to a standard commercial six or eight cylinder engine block within the capacity range specified in 3.5.1. After boring one bearing with power provided by the handcrank, a 1/2 inch electric drill shall be used for the remainder of the test until all cam and main bearings in the block have been bored, faced, and filleted. During the test, the boring machine shall be examined for compliance with 3.2, 3.5.1, and 3.5.2.

4.6.2 Accuracy test. When the test specified in 4.6.1 has been completed, the inside diameter, surface finish, and in-line accuracies of the bearings shall be measured to determine compliance with 3.5.3.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation and for Levels A and B packing and marking shall be in accordance with MIL-M-18058.

* 5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

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6.1 Intended use. This specification covers a universal boring machine that is portable and used to line bore, face, and fillet cam and main bearings in motor vehicle engine blocks. The machine is powered by either a hand operated crank that is furnished with the machine or by a 1/2 inch (or larger) electric drill that is not furnished.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Measuring and indicating device graduation(s) required (see 3.2.2).
(d) Exceptions and additional safety/health requirements when required (see 3.2.7 and 6.4).
(e) Painting and finishing if different (see 3.4.5).
(f) Carbide tool bits if different (see 3.6.3).
(g) First article inspection if different (see 4.3).
(h) Quality conformance inspection if different (see 4.4).
(i) Sampling if different (see 4.4.1).
(j) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

* 6.3.1 Repair part manuals. Repair part and accessory identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298. (See 3.2.3.)

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

* 6.5 Standard lubricants. The machine designer should specify and use, to the maximum extent possible, those lubricants (see 3.2.4) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 can be used as selection guides.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any

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